

After two decades of uninterrupted expansion the Morocco economy is facing several challenges. Economic structural reforms launched in the late 1990s led to increase the country's real gross domestic product (GDP) by nearly 120 percent between 2000 and 2019, expand per capita income by 72 percent, and almost eradicate extreme poverty. However, real GDP fell by 6.3 percent in 2020 – according to recent estimates – due to the COVID-19 pandemic, the largest contraction on record. After the gradual lifting of restrictions, economic activity began to pick up in late 2020, and Morocco's real GDP grew by 7.9 percent in 2021. This was supported by a successful vaccination campaign, a series of measures adopted to respond to the multifaceted nature of the shock, an extraordinary agricultural season, solid manufacturing and agro-industrial export, supportive counter-cyclical fiscal and monetary measures, and an unprecedented level of remittances. GDP growth dropped to just 1.2 percent in 2022,¹ resulting from overlapping shocks, including the sharp contraction in agricultural GDP due to the drought, the commodity price shocks triggered by Russia's invasion of Ukraine, a slowing global economy, and higher global energy and food prices. In response to the drought, in 2022 the Government of Morocco (GoM) adopted a US\$1 billion emergency recovery package for farmers, including agricultural insurance and rescheduling debt mechanisms. Economic growth is expected to accelerate to 2.5 percent in 2023.

The economic growth model that served Morocco well in recent decades shows signs of slowing down and needing to adapt to changing realities. More than half of GDP growth in recent decades was driven by fixed capital accumulation, which surpassed 30 percent of GDP since the early 2000s with two-thirds originating from public sources. The economy has not generated enough jobs to absorb a growing working-age population and overall unemployment remains high at around 12.8 percent in 2022, with urban youth unemployment reaching 33.4 percent.

Morocco's New Development Model (NDM) presents a new vision for the country's development.¹ The structural reforms launched two decades ago gave way to a sustained period of economic growth and poverty reduction that is unparalleled in the country's contemporary history. But, even before the abrupt recession triggered by the pandemic, the model exhibited signs of exhaustion, prompting the launch of an inclusive dialogue on how to reinvigorate Morocco's path toward sustained economic growth and social inclusion. This dialogue resulted in 2021 in the NDM, which sets ambitious development targets with a 2035 horizon, including doubling per capita GDP levels.

In his speech, His Majesty King Mohammed VI outlined four main guiding principles for water resources management in the context of increasing water deficits: (a) launching more ambitious programs and initiatives, and leveraging modern technology, particularly for water savings and wastewater reuse; (b) improving the rational exploitation and preservation of groundwater resources, in particular by curtailing illegal pumping and well digging; (c) fostering cross-sectoral coordination and updating sectoral strategies on a continuous basis, in light of their pressure on water resources and the need to build a water secured Morocco; and, (d) considering the real cost of water, accounting for each stage of the water mobilization process, and promoting transparency and awareness about all the drivers of cost of water.

According to the Country Climate and Development Report for Morocco (CCDR),¹² the greatest potential for putting Morocco on a climate-resilient and low-carbon pathway lies in (a) tackling water scarcity and droughts, notably through the nexus of water and agriculture; (b) enhancing resilience to floods to preserve urban and coastal economies and livelihoods; and (c) decarbonizing the economy for a zero-net emission pathway by the 2050s.

Hotter and drier conditions are also expected to increase demand, with crop water requirements predicted to increase by up to 12 percent, increasing demand for irrigation and placing further demands on limited water resources. These changes in supply and demand could result in a deficit of between 4 to 7 BCM per year by 2050.¹⁹ Such constraints could significantly affect the Moroccan economy,²⁰ with water scarcity exacerbating competition among water users and potential causing a 6.5 percent decline in real GDP under extreme conditions.²¹

8. To address the country's water challenges, the GoM is implementing comprehensive large-scale water sector investment programs.

households is expected to increase by 50 percent (from 1.1 to 1.7 BCM per year) during the same period, with corresponding increases in effluent quantities. Expected industrial demand growth is similar (from 241 to 370 MCM per year), and water from tourism is expected to more than double by 2050 (from 33 to 106 MCM per year).²² Based on an estimated population growth of 1.5 percent per year and economic growth of 4 percent per year.

Water demand from uABHn

However, there is also recognition that increasing resilience to climate change requires ambitious policy reforms that reflect the value of water, increase the transparency of costs along the water value chain, balance competing demands, and incentivize more efficient and rational uses. The PNAEPI acknowledges the incremental cost of new water supplies and the need for strong policy reforms, establishing targets for reducing water-losses, including provisions for the reduction of dam sedimentation and reforestation of watersheds, incentives for water savings in agriculture, drinking water, tourism, and industry, along with the preservation of groundwater resources.

As part of the continued commitment to régionalisation avancée, the GoM plans to establish regional multiservice companies (Sociétés Régionales Multiservices, SRM) for energy and potable water distribution and wastewater collection and treatment, to substitute autonomous municipal public utilities (Régies).² 5

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The Strategy for Gender Mainstreaming in the Water Sector (2017)² 6 focuses on developing institutional capacity to promote gender equality in the sector at the policy and program level by increasing women's representation in decision-making and in technical

² 2 While desalinization represents less than 2 percent of all mobilized water resources, it provides a high level of water security, with 44 percent for potable water, 41 percent for irrigation, and 14 percent to phosphate production.

Financial sustainability is further undermined by the under-collection of raw water abstraction and water pollution fees.² 8 The lack of a robust financing strategy for the sector could present barriers to the realization of the GoM's ambitious plan for expanding desalination capacity, particularly for irrigation purposes, and compromise the sector's financial viability with impacts on future water security.

In this context, an enhanced governance model is needed to ensure an integrated approach to water production and distribution, optimize allocation decisions, ensure financial sustainability, and ensure that decisions about water are made to maximize socio-economic outcomes in alignment with the development model for Morocco.

The Program for Results (the Program) aligns with the World Bank's twin goals of ending extreme poverty and promoting shared prosperity within the context of the World Bank Group's Country Partnership Framework (CPF) for Morocco FY19-24.³ 1 The CPF aims to contribute to social cohesion by improving the conditions for growth and job creation and reducing social and territorial disparities.

² 9 Desalination costs have dropped from a range of US\$2.5 to US\$44 per cubic meter in the 1980s to now less than US\$1.5 per cubic meter, and in many locations have fallen as low as around US\$0.5 to US\$0.6 per cubic meter. In addition to technological advances and economies of scale achieved by larger plants, falling costs are driven by project development choices, such as colocation of desalination plants with power plants, and enhanced competitiveness from more efficient methods of project financing and delivery.

The Program focuses on a portion of the PNAEP where the Government aims at enhancing efficiency, effectiveness, and impact of expenditures by linking the disbursement of funds to the achievement of specific results based on the existing performance-based budget system.

The Program will provide a platform for national and local entities to incentivize sustained progress of the government reform program, including on water valuation, state-owned enterprises, and the régionalisation avancée.

Based on the orientations provided by His Majesty King Mohamed VI, in 2017, a technical commission - under the guidance of the Interministerial Water Commission - was charged with the preparation of a program to accelerate investments in the water sector to strengthen drinking water supply and irrigation, in particular for the areas most impacted by the successive droughts impacting the country since 2015.⁸ With investments estimated at MAD 143 billion (US\$ 14.4 billion), the PNAEPI is expected to be financed through a combination of public subsidies, user contributions, and a combination of the private sector, climate financing, and international financing institutions.

The PNAEPI aims to diversify the sources of supply, guarantee water security, and reduce climate change impacts by accelerating investments to strengthen water supply for drinking and irrigation uses. The PNAEPI's major objectives are: (a) expanding water supply by increasing water storage capacity and the contribution of non-conventional sources to the water matrix (wastewater reuse and desalination), and safeguarding groundwater resources; (b) improving water efficiency by reducing water losses in conveyance and distribution networks (potable water and irrigation) and improving water productivity in the irrigated agriculture sector; (c) strengthening access to potable water supply in rural areas; and (d) increasing awareness on the value of water.

The Program will support interventions that address the water challenges on three levels: (a) sector-level governance focused on water planning, strengthening of groundwater management, basin-level capacity to manage water resources, data and information management systems; and performance of water and wastewater operators; (b) improvement of the financing sustainability of the sector, including through local-level improvements to improve water use efficiency and reduce non-revenue water; and, (c) promote non-conventional water, from the regulatory perspectives and by supporting investment to scale-up wastewater reuse.

The scope and boundaries of the Program are based on the activities identified in the PNAEPI and defined by priorities that contribute to the following: (a) strengthening the governance of the water sector; (b) improving the valuation and financial sustainability of the water sector; (c) creating a water-savings culture and reducing water losses; and (d) providing an enabling environment for the integration of non-conventional water resources.

This approach is built on: (a) strengthening the water sector governance; (b) improving financial sustainability and water use efficiency, and (c) enabling the integration of non-conventional water resource

The PDO is to strengthen water sector institutions and increase water availability in selected areas in Morocco.

These tools will facilitate GoM decision-making to bridge the gap between cost and revenues along the water value chain (including water mobilization through wastewater reuse and disposal), particularly considering significant transformation of the water matrix envisaged under the PNAE

This DLI supports the GoM's ambition to scale up treated wastewater for productive uses (green spaces, golf, industry, tourism, agriculture, and others) as a strategic resource to mitigate increasing water scarcity risks.

In addition, the DLIs will incentivize the implementation of treated wastewater reuse projects, including expanding the installed capacity to the treatment level required and the distribution network from the wastewater treatment plant to the taker of the treated effluent.

The Bank is supporting the preparation of a Public Finance Review in Morocco, including a deep dive into public expenditure analysis in the water sector.

The PNAEPI recognized the need to respond to the challenges of climate change and contribute to a green transition with a clear set of investments and accompanying measures that are critical for ensuring success.

The Program will also directly support the emissions reduction target in the NDC through investments in wastewater treatment and the energy efficiency gains that can be realized through investments in water loss reduction in distribution networks.

Program expenditures are estimated at MAD 5,801 million (equivalent to US\$ 573 million) and will finance activities at the central level by the MEE and the MoI, and at the local level by the ABHs and Régies

In 2020, the COVID-19 pandemic, coupled with a poor agricultural harvest due to drought, led to an increase in the budget deficit and public debt, which reached 7 percent and 67 percent of GDP -respectively.

Morocco's external debt experienced a significant increase at the end of 2022, to nearly 10 percent of GDP (MAD 229 billion; US\$ 21.7 billion). The budget deficit is MAD 73.1 billion against an initial forecast of MAD 72.6 billion.^{4 1} The MEF revised its budget deficit and growth forecasts for 2023 to 5.3 percent and 1.5 percent of GDP, respectively.

An evaluation of the costs and benefits confirms the economic rationale for the Program.^{4 2} The Program's economic feasibility analysis compared its estimated economic benefits and costs. The cost-benefit analysis estimated the economic feasibility of the Program by calculating the net present value (NPV) of cost and benefit streams and determining the Program ERR.

^{4 2} The Cost-benefit analysis is a method for comparing the economic pros and cons of policies and Programs to help policymakers identify the best or most valuable options to pursue. Cost-benefit analysis monetizes all major benefits and all costs associated with a project so that they can be directly compared with each other as well as to reasonable alternatives to the proposed project.

Benefits include (a) increased water revenues resulting from physical and commercial loss reductions; (b) decreased operational costs resulting from leak detection and reduced energy requirements; (c) revenue from the sale of treated wastewater; (d) GHG emission reductions owing to reduced energy requirements, and (e) reductions in local pollutants from WWTP effluents.

The economic analysis yielded an NPV of US\$231.8 million and ERR of 15.5 percent inclusive of the shadow price of carbon.⁴ 3
52.

A reduction in localized pollutants is expected from WWTPs upgrades to increase the capacity for tertiary treatment and reuse of treated effluent for green spaces and industrial and agricultural uses.

The Investments in Morocco CCDD estimated that a reduction in water availability could reduce GDP by up to 6.5 percent. water loss reduction will postpone the need to invest in non-conventional water sources.

In addition to the interventions of the Program, which are expected to increase revenues and reduce operational costs, a financial model of the water sector will be developed to improve financial sustainability and inform policy decisions. It will allow multiple institutions (ABHs, Régies, and water supply operators, among others) to improve the overall financial efficiency of the water sector and public expenditure management by combining capital expenditure planning with water pricing and ongoing operations.

The objective of the assessment was to examine whether Program systems provide reasonable assurance that the financing proceeds will be used for the intended purposes, with due attention to the principles of economy, efficiency, effectiveness, transparency, and accountability.

The internal controls of the implementing entities' fiduciary systems provide reasonable assurance that financing proceeds will be used to support the achievement of Program objectives, with due attention to the principles of economy, efficiency, effectiveness, transparency, and accountability.

The PPD includes several enhancements, such as introducing rated criteria, the selection of the most advantageous bid, competitive dialogue, and establishing an observatory for public procurement. Implementing the PPD will result in higher value for money with adequate levels of transparency, competitiveness, efficiency, and fairness by all Program entities.

Bank operations in the sector linked to water and irrigation, such as the Morocco Rural Water Supply Project and the Morocco Large Scale Irrigation Modernization Project, have faced challenges, with an extension of the closing date required to complete works and last payments.

The PforR instrument provides an opportunity to strengthen the transparency and predictability of resources, and budget execution, including procurement planning and execution, contract management, financial reporting, and anti-corruption mechanisms.

The Program is not expected to procure any large contracts valued at or above the Operational Procurement Review Committee (OPRC) thresholds (US\$75 million for works, US\$50 million for goods and non-consulting services, and US\$20 million for consultant services), which are based on a 'Substantial' risk rating.

The extension and rehabilitation of potable water distribution systems and the WWTPs upgrades will positively benefit direct and indirect job creation during the work phase. They will induce demand for goods and services that benefit local, regional, or national companies.

Potential social risks are mainly related to rehabilitating or constructing water supply and wastewater systems.

A further downturn in the global economic outlook could reduce exports, tourism receipts, and foreign direct investment.

The New Development Model and the reform agenda launched by the GoM would improve the attractiveness of Morocco for private investment, especially in promising sectors with high value-added, and mitigate the impact of these potential risks.

The increasing investment in non-conventional water resources, particularly desalination, is also changing the sector's cost structure.

The Program is expected to bring significant E&S benefits by strengthening sector institutions and increasing water availability.

The financial model shall capture CAPEX and OPEX and the different revenues, including national transfers and local contributions, taxes, and users' contributions (tariffs and fees)-.

This was due primarily to non-climatic factors, such as population growth in the north, the expansion of irrigation, and urban, industrial, and tourism development, with water demand already exceeding available supply. Based on the currently available sources of supply in the water accounts (i.e., freshwater, treated wastewater reuse, and desalinization), the annual average inflow is estimated at around 18 billion cubic meters (BCM) per year, with around 14.5 BCM per year available and sustainably exploitable with existing infrastructure, including 4 BCM per year of renewable groundwater resources.

Between 1960 and 2020, the Kingdom built more than 120 large dams, a tenfold increase in total water storage capacity, bringing the total to 135 large dams with a total capacity of 17.5 BCM.

Since 1995, the GoM has made significant investments to address the gap through the development of desalination plants, with the installed capacity reaching 220 MCM per year by the end of 2022. While desalinization represents less than 2 percent of all mobilized water resources, it provides a high level of assurance, with 44 percent for potable water, 41 percent for irrigation, and 14 percent for phosphate production.

Water
demand from urban households is expected to increase by 50 percent (from 1.1 to 1.7 BCM per year) during the same period, with corresponding increases in effluent quantities.

⁵ 7 A permanent reduction in water supplies could significantly affect the Moroccan economy, ⁵ 8 with water scarcity exacerbating competition among water users and leading to a 6.5 percent decline in real GDP under extreme conditions. ⁵ 9 For example, hotter and drier conditions are expected to increase the water requirement of crops by up to 12 percent, increasing demand for irrigation and further stressing limited water resources. While infrastructure will be required to help adapt to the changing conditions, the potential magnitude and uncertainty associated with future climate conditions will also require ambitious policy reforms that reflect the value of water resources, increase the transparency of costs along the water value chain, incentivize more efficient and rational uses, and balance competing demands.

The GoM is implementing comprehensive large-scale water sector investment programs to address the country's water challenges.

The PNAEPI promotes supply-side solutions, with a significant portion of investments aimed at increasing the mobilization of water resources by about 3 BCM per year by 2027, with a strong emphasis on non-conventional water resources. The PNAEPI establishes targets for reducing water-losses, acknowledging the incremental cost of new water supplies, and includes provisions for the reduction of dam sedimentation, incentives for

⁵ 2 Ministry of Equipment and Water, 2019.

⁵ 4 The implementation of the Irrigation Water Conservation Program 2008-2020 resulted in the conversion of 462,000 hectares to localized irrigation (395,000 hectares for individual reconversion and 67,000 hectares for reconversion of collective land (including ongoing works), intensifying agricultural production through more reliable irrigation and by doubling water productivity on average (measure as US\$/m³). ³ For instance, in the Doukkala ORMVA, the program contributed to increasing water productivity, from 5.7 to 10.6 MAD/m³ in 2018; and, to increase the share of high-value-added crops (vegetables, fruit orchards, and oilseeds) relative to low-value-added crops (cereals, beets, and fodder).

Regional Agricultural Development Offices (ORMVAs) provide irrigation services within a defined irrigated perimeter (usually encompassing large-irrigation schemes), with irrigation covering roughly 1.7 million hectares (accounting for 19.4 percent of the cultivated area), representing about 45 percent of the agriculture value, and contributing 75 percent of the total agriculture exports of the country.

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Mol, through its Direction of Public Local Networks (Mol-DRPL), plays a supporting role in planning, implementing, and monitoring the distribution of potable water and wastewater collection, which are the responsibility of the municipalities, exercise control over private concessions, and set retail tariffs for water and wastewater services.

⁶ 6 Evaluating the contribution of irrigation to economic and agricultural growth: the case of Morocco.

Low water tariffs could present barriers to realizing the GoM's efforts to boost water security, including further development of water supply sources (dams and water transfers) and the scale-up of non-conventional water resources (desalination and wastewater reuse).

These include support for the GoM's efforts to strengthen the governance of the water sector to ensure that it is fit for purpose and can deliver on Morocco's expectations related to the

⁶ 8 Source: Cour des Comptes

⁶ 9 Desalination costs have dropped from a range of US\$2.5 to US\$44 per cubic meter in the 1980s to now less than US\$1.5 per cubic meter, and in many locations have fallen as low as around US\$0.5 to US\$0.6 per cubic meter.

Also, given the ongoing reform on the distribution of water services, the Program provides an opportunity to support the improvement in performance benchmarking of public-owned service providers for electricity and water supply distribution and wastewater collection and treatment.

The impact of drought and floods and the pressure of growing demand are all factors that require planning for water resources development and management strategies. Thus, since the 1970s, the GoM has undertaken efforts for water resources planning aimed at meeting Morocco's water needs, with the objectives of (a) integrated planning and conjunctive management of surface and groundwater; (b) optimizing the allocation of water resources to meet present and future demand in the medium and long term in support of Morocco's economic and social development growth at national and local levels and also between sectors; (c) providing access to water for the various regions of the country to ensure balanced development and the promotion of water-poor regions through water transfers from surplus to deficit regions; and, (d) protecting water resources.

The PDAIREs include (a) a quantitative and qualitative assessment of water resources; (b) water demand by sector and use category and water balances and water resources development schemes; (c) the definition of the investments at the river basin; (d) the economic and environmental evaluation of the proposed schemes/investments; and, (e) the implementation modalities of these water resources development schemes/investments.

The Organic Law 113-1 (201) specifies that the municipality creates and manages the public services and facilities necessary for the offer of local services in the following areas: · the distribution of water and electricity; public lighting; · liquid and solid sanitation and wastewater treatment plants (Article 83).

They coordinate at the national level the plans for developing municipal public services, including their pricing.

There is no assessment of the available groundwater reserves, a projection of declining production, or an estimate of the cost increase of pumping and quality deterioration for overused groundwater.

The overexploitation of groundwater leads to additional costs for pumping and deepening wells/boreholes, which is problematic for small farmers who practice subsistence agriculture.

These challenges and changes in the sector's cost structure call for a comprehensive approach to the valuation of water that can inform future-focused policies and regulations to improve financial sustainability. These policies and regulations will be important to reveal the range of values among different stakeholders and develop a shared vision within communities, to optimize the sustainable and efficient allocation of water, ensure the equitable pricing of water services, and guide investments aimed at safeguarding water security, its contribution to economic growth and development, and maintaining social equilibrium.

The Program will build on these principles to identify and update the underlying costs and benefits associated with investments in the water sector to inform the development of a consolidated financial model.

This information is critical to informing and guiding the process of developing pricing strategies for the various sub-sectors that can be used to create incentives for better water use and consumption while safeguarding water for the environment and other intangible value allocation options.

Given Morocco's water scarcity challenges and the increasing cost of water mobilization, a further increase in potable water distribution efficiency is critical to enhance water security and support the financial sustainability of service providers.

This

would create additional water supply and allow new customers to connect to the supply system.

When a commercial loss is detected and resolved, the saving will be an immediate revenue increase and thus is based on the water sales tariff, generating additional revenues, and improving the financial performance of the beneficiary service providers.

As indicated by the NDM report and the PNAEPI, given the high-CAPEX costs associated with seawater desalination, mobilizing private-sector financing is key to Meet Morocco's water security objectives.

PPP contract models allow for the assessment of asset performance (desalinated water production capacity, quality, and water pressure at the point of delivery) against agreed-upon technical specifications, timing, and budget with the private party.

In 2016, 47 percent of desalination capital investments in the world (US\$2.9 billion) were financed using PPP schemes, with the MENA region leading with more than 30 desalination BOTs signed in the past 20 years.^{7 5}

Wastewater reuse scale-up

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In 2006, the government of Morocco embarked on a large National Sanitation Program (Programme National d'Assainissement Liquid et d'Épuration des Eaux Usées, PNA), which resulted in about 84 percent of the Moroccan urban population connected to a sewerage network in 2019 and increased the level of wastewater treatment from 8 percent in 2005 to 53 percent in 2019.

The PNAM, part of the PNAEPI, establishes that further developing and expanding treated wastewater reuse is a strategic solution to Morocco's water scarcity challenges. The PNAEPI foresees a total investment of MAD 3019 million (US\$301 million) to expand treatment capacity and upgrade WWTPs from primary or secondary treatment to tertiary treatment for reuse. The proposed investments build on Morocco's successful experience implementing reuse projects, for example, for use in industry, green spaces, and golf courses across the country.

infrastructure facilities across Morocco.

The company operates numerous mines, chemical plants, and

As part of its Sustainability Strategy, the OCP will expand its solar power grid as well as engage in energy efficiency practices to reduce consumption by 10 percent.

The scope of each reuse project will be specified in its corresponding reuse agreement, and it will include: the upgrading of existing wastewater treatment plants from primary or secondary treatment to tertiary treatment, the extension of tertiary treatment capacity, and the construction of treated wastewater distribution network and associated infrastructure (pumping stations, reservoirs, etc.).

It shall encompass dynamic planning that reflects the increasing uncertainty derived from climate change; and the principles to strengthen the sector's governance, institutional, and financial aspects (including defining water valuation and cost-recovery principles).

Also, the DLI provides incentives for signing participatory management contracts in three significantly overexploited aquifers (Berrechid, Maamoura, and Bahira).

This DLI provides incentives to strengthen the performance of the river basin agencies (ABHs), key institutions for water resource management at the basin level.

Rising temperatures and erratic rainfall have reduced river flows and increased evaporation and siltation of storage dams, leading to a 20 percent reduction in overall water resources in the last 30 years, exacerbating the problem of water scarcity generated by non-climate stressors such as population growth in the north, irrigation expansion, as well as urban, industrial and tourism development.

In 2020, the COVID-19 pandemic increased the budget deficit and public debt, reaching 7 percent and 67 percent of GDP, respectively. In 2020, the country's economic growth was strongly impacted by the health crisis and a poor harvest due to drought. In Morocco, the overall public debt ratio rose from 80.4 percent of GDP in 2019 to 94 percent in 2020, although it should not exceed 60 percent. Also, Morocco's external debt should experience a significant increase at the end of 2022, estimated at nearly 10 percent, to reach MAD 229 billion (US\$ 21.71 billion equivalent).

The MEF had revised its budget deficit and growth forecasts for the current year to 5.3 percent of GDP and 1.5 percent, respectively.

In the medium to long term, however, economic performance is expected to improve thanks to sound fiscal and monetary policies, better governance and more coherent sector strategies, and an improved investment environment, all of which aim to support gradual gains in competitiveness.

The 2023 Finance Law reveals the continuity of the efforts started with the 2021 and 2022 finance laws to stimulate revenue mobilization while consolidating expenditure and redirecting it toward social and strategic sectors. The 2023 Finance Law provides for the continued construction of the ‘Social State’ and the strengthening of its pillars via concrete and unprecedented measures that will improve the living conditions of large segments of the population, in particular, the generalization of social protection to vulnerable and needy categories, the improvement of the purchasing power of the middle class and the healthcare offer, the promotion of public schools and the development of mechanisms for access to housing.

The Government will attach particular importance to financing large-scale structural reforms initiated in recent years, to mobilize more resources and ensure their sustainability. Thus, the government will ensure the mobilization of all possible budgetary margins and direct them towards sectors with a strong economic and social impact.

The Program's economic feasibility analysis compares its estimated economic benefits and costs.

The cost-benefit analysis estimates the economic feasibility of the Program by calculating the net present value (NPV) of cost and benefit streams and by determining the Program's economic rate of return (ERR).

The cost-benefit analysis included economic costs and benefits under RA2 and RA3.

The net benefit is the difference between the incremental benefits and the incremental costs of two scenarios: without and with the Program.

Results under RA 2 and RA 3 were evaluated, measuring the expected costs and benefits for the Program's lifetime, estimated at 20 years for the investments included under RA2 and RA3.

- Costs and benefits are expressed in constant prices as of 2023.
- Discount rate.

The expected economic benefits of these investments include (a) increased water revenues resulting from physical and commercial loss reductions; (b) decreased operational costs as a result of leak detection and reduced energy requirements; (c) avoided capital expenditures; (d) GHG emission reductions owing to reduced energy requirements.

Additional water sales arising from greater water availability derived from reduced water losses. Reducing water losses is a key priority of the PNAEPI to help address water scarcity in Morocco and generate financial returns for the operators.

Investments in non-revenue water (NRW) reductions will increase the water sold directly by the 11 Régies and reduce the bulk water purchases from ONEE. As a result of the project, 20,000,000 m³ of water savings are expected to be achieved over the project's life with an average sale price of 4 MAD/m³ (US\$ 0.61/m³), generating additional revenues of up to US\$8 million per year. Investing in NRW reductions has been shown to reduce future capital expenditures owing to the extension of the useful life of current assets and a reduction in the rate of asset replacement.

Incremental O&M Savings. The economic analysis considers incremental O&M savings from reducing production costs due to reduced pumping costs resulting in reduced energy consumption.

Water loss reductions are expected to reach 54,000 m³/day, leading to savings of US\$55 million in capital expenditure.

Investment costs of US\$ 192 million include capital expenditures under RA2 related to the implementation of water loss reduction plans, including the deployment of geographical information management systems and hydraulic models; meters deployment (bulk- and micro-meters), network sectorization and pressure control Program; leakage detection and rehabilitation campaigns.

RA3 investments for increased wastewater reuse volumes. The expected economic benefits include (a) GHG emission reductions owing to a reduction in local pollutants from WWTP effluents; and (b) revenue from the sale of treated wastewater.

Besides direct economic benefits, many other potential benefits are not factored into the cost-benefit analysis. This is either because estimating such benefits is difficult due to the lack of data or it is challenging to quantify the value of those benefits because they might not be financial or economic.

Supporting the enabling environment and regulatory framework that facilitates the scaling up of non-conventional water resources provides Morocco with a viable path to securing its water future.

The economic analysis yielded an economic NPV of US\$ 230 million and ERR of 15.5 percent exclusive of the shadow price of carbon and an economic NPV of US\$ 231.8 million and ERR of 15.5 percent inclusive of the shadow price of carbon.

Scenario	Economic Rate of Return (ERR) and Net Present Value (NPV)	
	ERR (%)	NPV @ 6% (US\$ million)
Without shadow price of carbon	15.5	230
With the shadow price of carbon	15.5	231.8

Source: Team Calculations, June 2023.

The objective of the assessment was to examine whether Program systems provide reasonable assurance that the financing proceeds will be used for their intended purposes, with due attention to the principles of value for money, economy, efficiency, effectiveness, transparency, and accountability.

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PPD aligns with the best practices of international economics, efficiency, fairness, and integrity.

⁸ 7 The new decree aims to consolidate the public procurement system to provide more clarity to economic actors and improve the business climate. It also seeks to open opportunities for innovation, for new procurement features and approaches such as competitive dialogue and for small businesses and self-employed individuals (self-entrepreneurs) and establish a national preference for domestic products.

(ii): Regies and ABHs: They receive subsidies to contribute to financing certain projects included in the programs resulting from the policies for developing the water management and liquid sanitation sectors.

The draft budgets are prepared by the Authority and presented to the DRPL of the Ministry of the Interior (Regies) and DGH-MEE (ABHs) for approval of the investment projects in terms of consistency with sector-wide policies, strategies, and programs for the management of water, electricity, and liquid sanitation services.

The overall value of the building and public works (BTP) market forecast Program for the Ministry and its affiliated establishments for 2023 exceeds 45 billion dirhams.

The country scored A for PI-1 of PEFA 2016, mainly due to the high execution rate of the national budget, as mentioned in the report.

The CNCP's main mission involves professionalizing public procurement, ensuring compliance with transparency, efficiency, economy, and fairness principles, handling procurement-related complaints from bidders, and treating requests for opinions from public buyers.

It will induce demand for goods and services that benefit local, regional, or national companies.

